Pupillometry: The Eyes Shed Fresh Light on the Mind

Recent studies provide promising methodological advances in the use of pupillometry as on-line measurement of cognitive processes and show that visual attention allocation, mind-wandering, mental imagery, and even rhyme expectations can influence the size of the human pupil.

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Ocular muscles reflexively control the amount of light reaching the retina by dilating or contracting the pupil of the eye from diameters between 3 and 9 mm [1,2]. Small changes on the order of <1 mm reflect cognitive effort [3], because of the link between ocular muscles and the norepinephrine neurotransmitter system controlled by the locus coeruleus in the brainstem [4,5]. The locus coeruleus plays an important role in attentional processes, enabling the use of pupillometry as a window to changes in the state of mind [6]. Several recent studies have significantly advanced pupil-tracking methods and have extended our understanding of this fascinating link between eye and mind. On a methodological level, equipment-specific biases in pupil size estimation can now be corrected [7] and visually-induced pupil size oscillations predict visual selectivity with high temporal resolution [8], thus augmenting the arsenal of attention researchers. On the cognitive level, the list of pupil-related variables grows ever longer: both mind-wandering [9] and the mere imagination of brightness [10] induce pupillary responses, and effects of poetry on the pupil [11] confirm that beauty remains firmly in the eye of the beholder.

Correcting Biases in Pupil Size Estimation

One strength of pupillometry is that pupil-size information can nowadays easily be recorded by common eye-tracking devices without affecting participants' behaviour, for example during reading or scene perception. However, some methodological caution must be exercised when interpreting pupil size [7,12]. Brisson *et al.* [7] show that measurements depend on both gaze position and recording device: pupil size is underestimated for leftward and downward gaze, and overestimated for rightward and upward gaze when using an EyeLink© 1000 eye-tracker; the opposite result is obtained when using Tobii© (T120 and X120) eye-trackers.

These observations make it mandatory to counterbalance horizontal and vertical stimulus positions when interpreting pupil size in response to specific stimuli. Should this not be possible — for example in reading experiments where the beginning of a sentence is on the left side of the screen — then an interpretation of pupil size will be misleading. Fortunately, gaze-dependent biases can be corrected by means of the parameters reported by Brisson *et al.* [7].

Pupil Size Reflects Visual Attention

Naber *et al.* [8] introduce pupil frequency tagging as a powerful new method to track visual attention allocation on-line. In a first step, the authors established that repetitive changes in screen luminance (dark–bright–dark–bright and so on) are mirrored by pupil oscillations for frequencies up to 2.5 Hz. Next, they presented differently oscillating stimuli simultaneously on the screen and instructed observers to covertly attend to one (Figure 1A). A frequency spectrum analysis revealed that pupil oscillation power was selectively enhanced for the frequency of the attended stimulus (Figure 1B). These results show for the first time that directing attention to a stimulus increases pupil responses to that stimulus, indicating cortical contributions to pupil diameter.

On the neuronal level, Naber *et al.* [8] suggest that attention may facilitate the neural responses to temporal changes in luminance in early sensory brain areas which are linked to the brain stem nuclei controlling ocular muscles. Given this link, might it become possible to predict visual task performance from pupil behavior? Yes indeed! With a visual detection task, the authors go on to show that pupil oscillation power as long as 500 ms before target appearance predicts detection performance. In contrast, raw pupil diameter data were only predictive from 250 ms before target onset. These results recommend pupil frequency tagging, with its non-invasive nature and higher temporal resolution compared to standard arousal-based pupillometry, as a practical tool for further studies of visuo-spatial attention.

Pupillometry Predicts Mind-wandering

Another recent trailblazing study [9] used pupillometry to track the degree of task engagement. While reading a Sherlock Holmes story, participants in the study by Franklin *et al.* [9] were occasionally asked whether they had just been mind-wandering. Pupil size prior to questioning predicted their response, with larger size when participants were not reading for meaning.

This finding seems inconsistent with the typically reported pupil size increase under cognitive load [3–5], but it is in line with the inverse relationship between baseline and event-related pupil dilation. Accordingly, task engagement is associated with phasic activation of the locus coeruleus, resulting in lower baseline and higher event-related pupil dilation, whereas exploration for alternative tasks (mind-wandering) is associated with tonic activation of the locus coeruleus, resulting in higher baseline and lower event-related pupil dilation [13,14].

Pupil Responses in Relation to Imagined Light

Pupil responses to luminance are not purely stimulus-driven: the pupil also responds to illusions of brightness [15], pointing to a modulating role of expectations and contextual factors. Strikingly, Laeng and Sulutvedt [10] now report changes in pupil size when their participants merely imagined stimuli of different luminance: their pupils contracted when thinking about the sun and dilated when imagining a dark room. The effect occurred both for previously learned artificial stimuli (retrieved from short-term memory) and for familiar scenes retrieved from long-term memory.

These results are especially intriguing given that it is not thought possible to control the size changes of one's pupil

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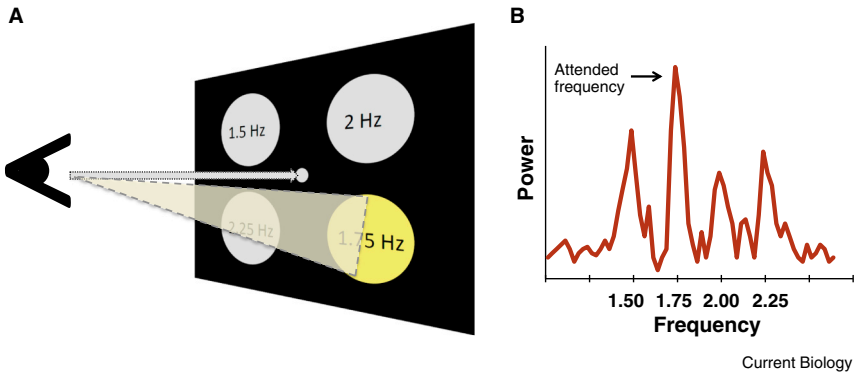


Figure 1. Illustration of Naber *et al.*'s procedure and results.

(A) Participants fixate a central dot (illustrated by the grey arrow) and direct their covert spatial attention to one of the four flickering stimuli (here to the 1.75 Hz stimulus, illustrated with the yellow beam). (B) Frequency spectrum analysis on the pupil oscillations reveals the highest power (i.e., strongest pupil oscillations) for the frequency of the attended stimulus.

at will [16]; the findings suggest that 'simulated' light is considered when our brain predicts the changes in pupil size necessary to protect the retina. Laeng and Sulutvedt's demonstration confirms that low-level features of a mental image induce neuronal responses similar to 'real' visual perception [17] and emphasize that higher level cognition can modulate lower level sensitivity.

From Rhyme to Reason: Pupillometry and Poetry

Aesthetic appreciation is among the highest echelons of cognitive performance, thought to encompass a wide range of subsidiary cognitive processes [18]. Scheepers *et al.* [11] presented their participants with auditory limericks, known for their rather predictable rhyme scheme (AABBA). Importantly, the last line of each limerick contained a semantic, syntactic, metric, or rhyme violation. Participants' pupils dilated much more to the rhyme violation compared to the others, confirming that the listeners had developed a very specific expectation about how each poem should end. The effect was present just 200 ms after the end of the critical word — the moment of rhyme violation — suggesting good temporal sensitivity of this method. Given that poetry captures our minds predominantly through emotional engagement, pupillometry is an ideal tool for further studies of poetic devices.

Outlook

The studies presented here highlight promising new applications of pupillometry in the cognitive sciences. The advances in temporal resolution of this technique, combined with its unobtrusiveness and practicality, enable it to compete with (or supersede) other methods currently used in attention research, such as the more effortful steady-state visually evoked potentials used in electrophysiological studies (EEG/MEG). Moreover, pupil frequency tagging could be used to capture the spatial extent of neglect or other attention-related disorders.

Importantly, these studies set the stage for the development of pupil-based assistive feedback. For example, pupil feedback may be used to prevent mind wandering during car driving or CCTV monitoring. In a clinical setting, Stoll *et al.* [19] recently supported communication with locked-in syndrome patients by means of pupil responses. This new on-line account to the mind may also help reduce the "noise" in behavioral measures. But among all these new promises, researchers should also note the cognitive implication of eye aperture itself: merely squinting or gaping with the eyes will quickly bias cognition towards better discrimination or better detection, respectively [20]. Thus, the eyes turn out to be an ever more complex set of filters for the mind.

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http://dx.doi.org/10.1016/j.cub.2014.02.028